BRD

1. Business Requirements

1.1 Problem Statement

Neurological diseases affect 1 billion+ people globally. Current diagnosis requires specialist visits (6-18 month wait), expensive imaging ($2K-5K), and lacks AI governance frameworks for clinical deployment.

1.2 Objectives

Objective	Target	Metric
Diagnostic accuracy	≥ 95% all diseases	CV Accuracy, F1, AUC
Time to diagnosis	< 5 seconds	Inference latency
Cost reduction	$300-800 wearable	Cost per screening
Regulatory compliance	FDA, ISO, HIPAA	Compliance score
Explainability	Clinician + patient	SHAP, RAG faithfulness
Governance	97 audit dimensions	RAI compliance 0.91

1.3 Stakeholders

Stakeholder	Role	Concern
Clinicians / Neurologists	End users	Accuracy, explainability, override
Patients / Caregivers	Beneficiaries	Accessibility, privacy, plain-language
Regulatory Bodies	Compliance	SaMD classification, validation
Data Scientists	Builders	Performance, reproducibility
AI Governance Officers	Oversight	Fairness, bias, audit trails
Hospital Administration	Procurement	ROI, integration, liability

1.4 Disease Performance

Disease	CV Acc	Ext Acc	F1	Risk
Epilepsy	100.00%	100.00%	1.000	LOW (15)
Parkinson's	100.00%	100.00%	1.000	LOW (15)
Alzheimer's	100.00%	100.00%	1.000	LOW (15)
Schizophrenia	100.00%	100.00%	1.000	LOW (10)
Depression	100.00%	100.00%	1.000	LOW (15)
Autism	96.84%	97.50%	0.970	LOW (32)
Stress	100.00%	100.00%	1.000	LOW (10)

HLD

2. High-Level Design

2.1 System Context (C4 Level 1)

```
Patient / Clinician
        |
   EEG Signal + Clinical Context
        |
        v
+---------------------------------------------+
|     NeuroMCP-Agent v2.5.0 (RGAIG+)         |
|  7 disease detection | Multi-agent         |
|  RAG explanations | 46 RAI modules         |
|  97 audit dimensions | MCP/A2A protocol    |
+---------------------------------------------+
     |              |              |
     v              v              v
 EEG Devices    Medical KB     Regulatory
 (Emotiv)      (ChromaDB)     Standards
```

2.2 Container Diagram (C4 Level 2)

```
+---------------------------------------------------------------+
|                 NeuroMCP-Agent System                          |
|                                                               |
|  Streamlit UI    MCP Server       REST API (FastAPI)          |
|  (12 tabs)       (JSON-RPC 2.0)                               |
|       |               |               |                       |
|       +-------+-------+-------+-------+                       |
|               |                                               |
|               v                                               |
|  +---------------------------------------------------------+  |
|  |          Agentic AI Orchestrator                         |  |
|  |  Coordinator | Validator | Governor                      |  |
|  +---------------------------------------------------------+  |
|               |                                               |
|  +---------------------------------------------------------+  |
|  |        A2A Message Bus (JSON-RPC 2.0, Pub/Sub)         |  |
|  +---------------------------------------------------------+  |
|    |     |     |     |     |     |     |                    |
|   Epi   PD   Alz   ASD  Sch   Dep  Stress                   |
|  Agent Agent Agent Agent Agent Agent Agent                    |
|               |                                               |
|  +---------------------------------------------------------+  |
|  |  EEG Pipeline | Ultra Stacking | RAG | RAI (46 mod)   |  |
|  +---------------------------------------------------------+  |
```

```
|  +-----------------------------------------------------+  |
|  |  ChromaDB (1.4GB) | Models (198) | EEG Datasets       |  |
|  +-----------------------------------------------------+  |
+-----------------------------------------------------------+
```

2.3 Technology Stack

Layer	Technology	Purpose
Frontend	Streamlit	Dashboard (12 tabs)
API	FastAPI + MCP	REST + JSON-RPC 2.0
Agents	Custom Python	7 disease + 3 orchestration
ML/DL	sklearn, XGBoost, LightGBM	Ultra Stacking Ensemble
RAG	ChromaDB + Sentence-BERT	Retrieval-Augmented Generation
RAI	46 custom modules	1300+ analysis types
Wearable	Emotiv SDK	Insight / EPOC X / EPOC Flex

LLD

3. Low-Level Design

3.1 EEG Preprocessing Pipeline

```
RAW EEG --> Signal Loading --> Segmentation (2-4s @ 256Hz)
  --> Feature Extraction (47 features) --> NaN Handling
  --> Train/Test Split (5-Fold) --> Augmentation (50->200)
  --> Standardization --> Ultra Stacking Ensemble (15 clf)
```

3.2 Feature Categories (47 Total)

Category	Count	Examples
Statistical	15	Mean, std, skewness, kurtosis, RMS
Spectral	18	Band powers (delta, theta, alpha, beta, gamma), PSD
Temporal	9	Hjorth parameters, zero crossings
Nonlinear	5	Approximate entropy, Hurst exponent

3.3 Ultra Stacking Ensemble (15 Classifers)

#	Classifier	Count	Parameters
1-2	ExtraTrees	2	n_est=200, max_depth=10
3-4	Random Forest	2	n_est=200, max_depth=10
5-6	Gradient Boosting	2	n_est=100, lr=0.1
7-8	XGBoost	2	n_est=100, max_depth=6
9-10	LightGBM	2	n_est=100, leaves=31
11-12	AdaBoost	2	n_est=100, lr=0.5
13-14	MLP	2	(256,128), dropout
15	SVM	1	kernel=rbf, C=1.0
Meta	MLP Meta-Learner	1	(256,128), dropout

3.4 MCP Server (12 Tools)

Tool	Description	I/O
classify_eeg	Classify EEG signal	signal - disease, confidence
extract_features	Extract 47 features	raw signal - feature vector
explain_prediction	SHAP explanation	prediction - feature contrib
validate_signal	Quality check	signal - quality score
get_rai_report	RAI analysis	model - compliance report
query_knowledge	RAG query	question - answer + citations
monitor_drift	Drift detection	new data - drift score

4. System Architecture

4.1 Three-Tier Agentic Architecture

```
TIER 1: Multi-Modal Data Preprocessor
  EEG loading | Band-pass filter | ICA | Segmentation | 47 features
                         |
                         v
TIER 2: Agentic Decision System
  Data type detection | Signal quality | Model routing | Consensus
                         |
                         v
TIER 3: Specialized Analysis Modules
  Disease agents (7) | Ensemble prediction | RAG explanation | RAI (46)
```

4.2 AI Governance Structure

```
              GOVERNANCE BOARD
     Policy | Ethics | Compliance | Risk | Audit
                     |
       +-------------+-------------+
       v             v             v
   ETHICAL AI     SAFE AI     SYMBIOTIC AI
   Fairness       Fail-safe    Human-AI
   Privacy        Bounded      Collaboration
   Autonomy       Monitored    Augmentation
   Dignity        Verified     Feedback
                     |
         MODEL CONTROL PORTAL (MCP)
   Registry | Versioning | Audit | Access | Rollback
```

5. Software Architecture

5.1 Agent Lifecycle

```
IDLE -> INITIALIZING -> READY -> PROCESSING -> READY
                                     |
                                  WAITING -> READY
                                     |
                                  ERROR -> TERMINATED
```

5.2 Disease Agent Capabilities

Agent	Capabilities	Output
AlzheimerAgent	MRI analysis, progression staging, biomarker assessment	Stage (CN/MCI/AD)
ParkinsonAgent	Gait analysis, tremor assessment, motor evaluation	PD score
EpilepsyAgent	Seizure detection, type classification, focus localization	Seizure type
AutismAgent	Social analysis, connectivity assessment	ASD probability
SchizophreniaAgent	Perception analysis, cognition assessment	Risk score
DepressionAgent	Mood analysis, affect assessment	Severity level
StressAgent	Arousal analysis, load assessment	Stress level

ADR

6. Architecture Decision Records

ADR	Decision	Rationale	Trade-off
ADR-001	Ultra Stacking (15 clf) over single model	99.55% avg accuracy via ensemble diversity	Higher latency (~2s), 380 MB total
ADR-002	MCP Protocol (JSON-RPC 2.0)	Standard tool discovery, agentic ecosystem	Custom server/client implementation
ADR-003	ChromaDB for RAG vector store	Lightweight, embeddable, no external deps	Single-node only
ADR-004	46-module RAI framework	No existing framework covers all dimensions	~2 MB codebase, maintenance overhead
ADR-005	Consumer EEG (Emotiv) devices	$300-800 vs $2K-5K, home-based screening	Lower SNR than clinical-grade
ADR-006	Bootstrap (1000x) over Monte Carlo	Non-parametric, distribution-free, small N	Simpler but valid for classification

C4

7. C4 Model Diagrams

7.1 Level 3: Component Diagram (Agentic Layer)

```
+-----------------------------------------------------------------+
|                     Agentic AI Layer                            |
|  +---------------------+                                        |
|  | Coordinator Agent   | Routes, manages workflow               |
|  +---------------------+                                        |
|  | Validator Agent     | Validates predictions, quality         |
|  +---------------------+                                        |
|  | Governor Agent      | Enforces RAI policies, audit           |
|  +---------------------+                                        |
|            |                                                    |
|  +-------------------------------------------------------+     |
|  | A2A MessageBus (JSON-RPC 2.0, Pub/Sub, mTLS, JWT)   |     |
|  +-------------------------------------------------------+     |
|    |    |    |    |    |    |    |                            |
|   Alz  PD  Epi  ASD Sch  Dep Stress                         |
|  +----+----+----+----+----+----+----+                           |
|  Disease-Specific Agents (specialized capabilities)            |
+-----------------------------------------------------------------+
```

8. Sequence Diagrams

8.1 Disease Detection Flow

```
Patient -> EEG Device -> Preprocessor -> Decision System -> Disease Agent
                                                                    |
                                                              Ultra Stacking
                                                              (15 classifiers)
                                                                    |
                                                              RAG Engine
                                                              (explanation)
                                                                    |
                                                              Clinician
                                                              (review/override)
```

8.2 Agent-to-Agent Communication

```
Coordinator --> MessageBus --> EpilepsyAgent (process)
                               --> ValidatorAgent (validate)
                               --> GovernorAgent (RAI check)
                           --- RESPONSE (diagnosis + RAI compliant)
```

8.3 Hallucination Detection

```
Query -> RAG Retriever -> LLM Generator -> Hallucination Detector
                                                          |
                                                  NLI + Entity + Claim
                                                          |
                                              [GROUNDED] --> Return with confidence
                                          [HALLUCINATION] --> Regenerate stricter
```

9. Analysis Types (1300+)

9.1 Core RAI (11 modules, 480+ types)

#	Module	Types	Frameworks
1	reliability_analysis	54	Reliable, Trustworthy AI
2	safety_analysis	38	Safe AI, Long-Term Risk
3	accountability_analysis	54	Accountable, Auditable
4	fairness_analysis	56	Fairness, Ethical, Social AI
5	privacy_analysis	38	Privacy-Preserving
6	interpretability_analysis	54	Interpretable, Explainable
7	human_ai_analysis	36	Human-Centered, HITL
8	lifecycle_analysis	38	Lifecycle Management
9	monitoring_analysis	58	Drift Detection, Debug
10	sustainability_analysis	38	Green AI, Environmental
11	generative_ai_analysis	18+	Responsible GenAI

9.2 Extended (10 modules, 180+ types)

#	Module	Types
12	energy_efficiency_analysis	18
13	hallucination_analysis	20
14	hypothesis_analysis	20
15	threat_analysis	20
16	swot_analysis	5+
17	governance_analysis	20
18	compliance_analysis	20

#	Module	Types
19	responsible_ai_analysis	20
20	explainability_analysis	20
21	security_analysis	20

9.3 Advanced + Governance + Trustworthy + Data (25 modules, 640+ types)

Group	Modules	Types
Research / Quality	fidelity, probability, divergence, human_evaluation	70+
Advanced Evaluation	evaluation_dimensions, text_relevancy, performance_governance, factual_consistency, diversity_creativity	125+
RAI Governance	rai_pillar, data_policy, validation_techniques, control_framework	110+
12-Pillar Trustworthy	portability, trust_calibration, lifecycle_governance, robustness_dimensions	125+
Master Data (v2.5)	data_lifecycle, model_internals, deep_learning, computer_vision, nlp, rag, ai_security	275+

Grand Total: 46 modules | 1300+ analysis types | 1105 Python exports

10. Statistical Analysis

10.1 Validation Methods

Method	Description
5-Fold Stratified CV	Primary validation, all 7 diseases
LOSO	Leave-One-Subject-Out, generalization
External Holdout (20%)	Independent test set
Bootstrap CI (1000x)	95% confidence intervals
McNemar's Test	Statistical significance (p < 0.05)
DeLong Test	AUC comparison (p < 0.05)

10.2 Bias Detection

Metric	Score	Threshold	Status
Demographic Parity	0.03	< 0.10	PASS
Equal Opportunity	0.05	< 0.10	PASS
Predictive Equality	0.04	< 0.10	PASS
Calibration Within Groups	0.97	> 0.90	PASS
Individual Fairness	0.92	> 0.85	PASS

10.3 RAI Compliance

Dimension	Score	Status
Fairness	0.92	PASS
Privacy	0.95	PASS
Safety	0.95	PASS
Transparency	0.88	PASS

Dimension	Score	Status
Robustness	0.85	PASS
Data Quality	0.94	PASS
Overall	**0.91**	COMPLIANT

11. 5-Pillar RAI Deep Audit (97 Dimensions)

Pillar	Dimensions	High Risk	Focus Areas
1. Data Responsibility & PHI	18	14 (78%)	PHI, De-ID, Encryption
2. Model Responsibility	19	14 (74%)	Fairness, XAI, HITL
3. Output & Clinical Safety	20	13 (65%)	Safety, Confidence, Harm
4. Monitoring & Drift	20	16 (80%)	Drift, IR, Rollback
5. Governance & Compliance	20	16 (80%)	Structure, Risk, Audit
TOTAL	97	73 (75%)	--

Regulatory Standards

Standard	Domain	Pillars
HIPAA	US Healthcare Privacy	1, 4, 5
FDA SaMD	Medical Device Software	2, 3, 5
ISO 14971	Medical Device Risk	3, 5
ISO/IEC 42001	AI Management System	5
GDPR	EU Data Protection	1, 5
NIST AI RMF	AI Risk Management	All

12. Testing Framework

Level	Scope	Tools	Coverage
Data Testing	Quality, drift, bias	Great Expectations, Deequ	100% pipelines
Model Testing	Unit, integration, perf	pytest, MLflow	95% code
Accuracy Testing	Metrics, benchmarks	sklearn, custom	Cross-validation
Business Testing	KPIs, ROI, clinical	Custom dashboards	All rules
Aspect Testing	Fairness, privacy, safety	Fairlearn, PySyft	All RAI

Literature Comparison

Study	Disease	Reported	Ours	Improvement
Andrzejak (2001)	Epilepsy	97.0%	100.0%	+3.0%
Ahmadlou (2012)	Alzheimer's	95.7%	100.0%	+4.3%
Bosl (2018)	Autism	81.0%	96.8%	+15.8%
Acharya (2015)	Epilepsy	98.0%	100.0%	+2.0%
Murugappan (2019)	Depression	93.2%	100.0%	+6.8%